

Jatin Shivaprakash

Software Engineer

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SUMMARY

Software Engineer with **3+ years of experience** in **backend development, distributed systems, cloud-native delivery, and AI-enabled applications**. Progressed from **REST APIs**, database workflows, testing, and production support at **DXC Technology** to **Kafka-driven services, governed RAG, and Agentic RAG** at **Thoughtworks**. Brings feature ownership, problem solving, and cross-functional collaboration across QA, platform, and engineering teams, delivering reliable backend systems, secure integrations, scalable services, and production-ready AI capabilities.

SKILLS

Languages & Frontend: Python, Java, Go, JavaScript, TypeScript, C++, SQL, React.js, HTML, CSS

Backend & APIs: FastAPI, Spring Boot, REST APIs, Pydantic, SQLAlchemy, OpenAPI/Swagger

AI & Retrieval: RAG, Agentic RAG, LangGraph, LangChain, Tool Calling, HITL, Hybrid Retrieval, RAGAS, NLP, scikit-learn, Pandas

Databases & Search: PostgreSQL, MongoDB, MySQL, Supabase, pgvector, PostGIS, Query Optimization

Distributed Systems & Messaging: Kafka, Redis, RabbitMQ, Celery, Distributed Systems, Event-Driven Architecture, Idempotency

Cloud, DevOps & Observability: AWS, Docker, Kubernetes, CI/CD, Jenkins, GitHub Actions, OpenTelemetry, CloudWatch

Testing & Security: pytest, Postman, API Testing, Integration Testing, Regression Testing, OAuth 2.0, JWT, RBAC

Tools & Engineering Practices: Git, Jira, Agile, Scrum, Pull Requests, Code Reviews, Root-Cause Analysis

EXPERIENCE

Thoughtworks, Software Engineer

Jun 2025 – Present | USA

- Build backend transaction services with **Python, FastAPI, PostgreSQL, Redis**, and **Kafka**, delivering **10+ REST endpoints** while owning **requirements clarification**, implementation, **testing**, integration, and **production validation** across client workflows.
- Develop governed **RAG** knowledge search with **LangChain, Supabase, pgvector, Hybrid Retrieval**, and **Reranking**, improving retrieval time by **15%** while enabling **permission-aware citations** and structured discovery for operations teams.
- Extend operations approval workflows with **Agentic RAG, LangGraph, Tool Calling, HITL**, and **RBAC**, integrating **5 governed actions** with retries, **Structured Outputs**, and auditable execution controls.
- Validate backend and **RAG** releases using **pytest, Postman, RAGAS**, and **Golden Datasets**, maintaining **66% automated coverage** across API behavior, retrieval quality, access controls, and regression checks.
- Deploy containerized services using **Docker, Kubernetes, AWS, CI/CD**, and **OpenTelemetry** across **3 environments**, partnering with platform engineers to troubleshoot releases, trace failures, and verify production health.
- Create AI-assisted release intelligence using **Jenkins, GitHub Actions, Structured Logging**, and operational telemetry, reducing build-failure triage time **14%** by correlating tests, deployment events, runbooks, and incidents.

DXC Technology, Software Engineer

Jun 2022 – Jul 2024 | India

- Developed assigned customer and service-request features using **Python, FastAPI, Pydantic, REST APIs**, and **Swagger**, contributing **8+ endpoints** while partnering with **Business Analysts** and **QA** to improve maintainability.
- Optimized **PostgreSQL** access for customer-history workflows using **SQLAlchemy, SQL, Alembic, indexing**, and **pagination**, reducing **API response latency by 18%** while improving transaction reliability across assigned modules.
- Implemented secure notification and synchronization services with **JWT, OAuth 2.0, RBAC, Redis, Celery**, and **RabbitMQ**, reducing repeated database requests by **14%** across routine application workflows.
- Expanded release confidence using **pytest, unittest, Postman, integration testing, regression testing, Git, pull requests**, and **code reviews**, reaching **78% automated coverage** before scheduled application releases.
- Supported containerized application releases through **Docker, Jenkins, Kubernetes, AWS**, and **CloudWatch** across **3 environments**, troubleshooting configuration issues and validating deployment health alongside **DevOps** engineers.
- Resolved production support issues using **logging, monitoring, root-cause analysis, Jira, Agile/Scrum**, and **CI/CD troubleshooting** while enhancing internal screens with **React, TypeScript, HTML**, and **CSS**.

PROJECTS

LedgerFlow | **Python, FastAPI, PostgreSQL, Docker**

- Built a payment ledger with **Python, FastAPI, PostgreSQL**, and **REST APIs**, implementing **double-entry accounting** across **3 core workflows**: transaction posting, balance tracking, and ledger history.
- Implemented **concurrency-safe posting** with **PostgreSQL row-level locking** and **idempotency keys**, then benchmarked the **Dockerized** service using **pytest** and **Locust**, sustaining **340 requests/second** with **56 ms p99 latency**.

Distributed Community Platform | **Go, ProtoActor, REST APIs**

- Engineered a **3-layer distributed platform** in **Go** with **ProtoActor** and **REST APIs**, separating API handling, actor messaging, and simulation workloads to improve **scalability** and maintainability.
- Designed and **load-tested REST APIs** for community interactions, using **actor-based concurrency** and **multi-process simulation** to evaluate responsiveness under **thousands of user activities** and validate distributed execution.

EDUCATION & CERTIFICATIONS

Master of Science in Computer Science, University of Florida

Aug 2024 – May 2026 | Gainesville, FL

- AWS Academy Cloud Operations [[Link](#)]
- Machine Learning – Coursera [[Link](#)]